

Project Final Presentation

H-Neurons in Small-Scale Models

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Motivation:

Gao et al. (2025) — arXiv:2512.01797

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Problem

- ▶ LLMs can give answers that sound reasonable, but are not true.
- ▶ In question answering, this is easy to see: an answer can be either factually correct or hallucinated.
- ▶ The paper argues that to understand this behavior, we should not only look at the final text, but also at what happens inside the model.

Reference finding

A very small subset of FFN neurons is strongly associated with hallucination:

$< 0.1\%$ of all FFN neurons

They also showed that these neurons:

- ▶ emerge during pre-training;
- ▶ increase hallucination when amplified;
- ▶ reduce hallucination when suppressed.

They called these neurons **H-Neurons**.

Q1 — Existence

Do H-Neurons exist even in small-scale models?

Gao et al. (2025)

- ▶ Smallest tested model: 7B parameters
- ▶ Evaluated on large LLMs only
- ▶ Models below 7B were not systematically analyzed

Q2 — Scale

Does H-Neuron-based classifier performance increase with model size?

Q3 — Generalization

Do H-Neurons identified on TriviaQA transfer to NQ-Open?

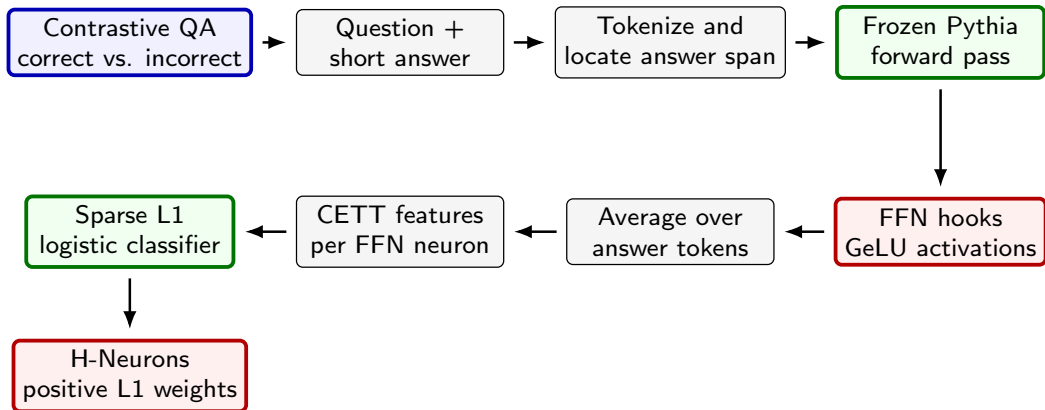
This project

- ▶ Uses the Pythia family
- ▶ Controlled training data and procedure
- ▶ Enables scale comparison from small to larger models

Goal: Answer the Q1, Q2 and Q3 for Pythia family.

Proposed Approach

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Only the sparse classifier is trained. The language model itself remains frozen.

In-domain dataset

- ▶ TriviaQA rc.nocontext.
- ▶ The model answers from memory, without context passages.
- ▶ The final dataset contains 100 balanced examples per model (50 correct + 50 incorrect)

Out-of-domain dataset

- ▶ NQ-Open validation split.
- ▶ The final dataset contains 30 balanced examples per model (15 correct + 15 incorrect)
- ▶ Used only for OOD evaluation; the L1 classifier is not retrained.

Model size	Sampling regime	Correctness threshold	Notes
< 1.4B parameters	$n = 1$	1/1	Multi-sampling infeasible
≥ 1.4 B parameters	$n = 10$	8/10	Early stopping after 3 consecutive failures

Few-shot prompt format

Question: what is the capital city of france Short answer: Paris

Question: when was the last time anyone was on the moon Short answer: December 1972

Question: who was the first woman to win a nobel prize Short answer: Marie Curie

Question: what planet is closest to the sun Short answer: Mercury

Question: In which decade did stereo records first go on sale? Short answer:

Setting	Model output	Evaluation
Zero-shot	The 1950s.	Incorrect
Few-shot	1950s	Incorrect

Models

- ▶ Pythia base models: 70M, 160M, 410M, 1B, 1.4B, 2.8B, 6.9B.
- ▶ GPT-NeoX decoder-only Transformer architecture.
- ▶ Models below 1.4B were loaded on CPU in float32; larger models used 8-bit GPU offload for the first 24 layers and float32 CPU for the remaining layers.

Tokenizer Setup

Parameter	Value
Maximum generated tokens	16
Sampling temperature	1.0
Top- k sampling	50
Top- p nucleus sampling	0.9
Sampling enabled	Yes
Max. response length (words)	10
Max. response length (characters)	100

Question

Which US President was born Lesley Lynch King Jr?

Response entering the model

Jimmy Carter

Full text sent to the model

Question: Which US President was born
Lesley Lynch King Jr?

Short answer: Jimmy Carter

Answer token indices

answer_start=17, answer_end=19

$$n_{\text{answer tokens}} = 2$$

Activation selection

During the forward pass, FFN activations are captured for all tokens.

Only answer-token positions are selected:

$$z_{\ell,t} \quad \text{for} \quad t \in [\text{answer_start}, \text{answer_end})$$

$$z_t = \sigma(W_{\text{gate}}x_t) \odot (W_{\text{up}}x_t), \quad x_t \in \mathbb{R}^d, \quad W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_m \times d}, \quad z_t \in \mathbb{R}^{d_m}.$$

Model	L	d	d_m	N_{FFN}
Pythia-70M	6	512	2,048	12,288
Pythia-160M	12	768	3,072	36,864
Pythia-410M	24	1,024	4,096	98,304
Pythia-1B	16	2,048	8,192	131,072
Pythia-1.4B	24	2,048	8,192	196,608
Pythia-2.8B	32	2,560	10,240	327,680
Pythia-6.9B	32	4,096	16,384	524,288

L : layers; d : hidden size; d_m : MLP width;

$$N_{\text{FFN}} = L \times d_m.$$

After averaging answer-token activations, each neuron has one value $\bar{z}_{i,\ell,j}$ for example i , layer ℓ , and neuron j .

$$\text{CETT}_{i,\ell,j} = \frac{|\bar{z}_{i,\ell,j}|}{\|\bar{z}_{i,\ell}\|_2 + \varepsilon}, \quad \|\bar{z}_{i,\ell}\|_2 = \sqrt{\sum_{j=1}^{d_m} \bar{z}_{i,\ell,j}^2}$$

CETT turns raw FFN activations into relative neuron-contribution features:

$$X_i = [\text{CETT}_{i,1,1}, \dots, \text{CETT}_{i,L,d_m}] \in \mathbb{R}^{N_{\text{FFN}}}$$

Hallucination probability

The classifier receives the CETT vector X_i , not the text.

$$s_i = b + X_i w = b + \sum_{j=1}^N X_{i,j} w_j$$

$$p_i = P(y_i = 1 \mid X_i) = \sigma(s_i) = \frac{1}{1 + e^{-s_i}}$$

$$y_i = 1 \implies \text{hallucinated answer}$$

Higher p_i means higher estimated hallucination probability.

Sparse neuron selection

The same classifier is trained with an L1 penalty:

$$\mathcal{L} = - \sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^N |w_j|$$

The L1 term pushes most weights to zero.

Non-zero weights select informative neurons; positive weights indicate neurons associated with hallucination.

$$\mathcal{H} = \{j : w_j > 0\}$$

Each classifier weight w_j corresponds to one global FFN neuron.

$$w_j > 0$$

increases the probability of
hallucination

$$w_j < 0$$

pushes the decision toward
correct answers

$$w_j = 0$$

neuron discarded by L1

Since $y = 1$ denotes hallucination, H-Neurons are defined as:

$$\mathcal{H} = \{j : w_j > 0\}$$

$$\%H = \frac{|\mathcal{H}|}{N_{FFN}} \times 100$$

Results: Accuracy and AUROC

AUROC is the probability that a hallucinated answer receives a higher score than a correct answer.

Table 1: Comparative H-Neuron results across the Pythia family. † indicates models with methodological limitations. Acc H: H-Neuron classifier accuracy; Acc Rand: random-neuron baseline accuracy; AUROC H: in-domain AUROC; AUROC OOD: NQ-Open ROC-AUC.

Model	H-Neur.	% Total	Acc H	Acc Rand	AUROC H	AUROC OOD
Pythia-70M†	11	0.0895%	0.85	0.6626	0.92	0.50
Pythia-160M	18	0.0488%	0.60	0.5596	0.71	0.72
Pythia-410M	22	0.0224%	0.75	0.5648	0.80	0.84
Pythia-1B†	3	0.0023%	0.75	0.5325	0.82	0.80
Pythia-1.4B	68	0.0346%	0.85	0.6049	0.90	0.84
Pythia-2.8B	105	0.0320%	0.90	0.5938	0.92	0.90
Pythia-6.9B	71	0.0135%	0.75	0.5951	0.85	0.80

H-neurons < 0.1% of all FFN neurons

H-Neurons vs. Random Neurons in the Pythia Family

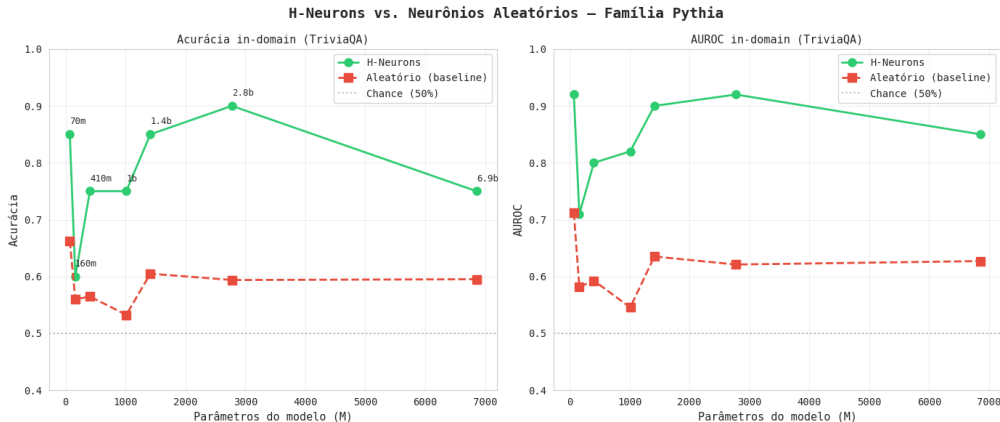


Figure 1. H-Neurons vs. random neurons baseline in the Pythia family. Left: in-domain accuracy on TriviaQA. Right: in-domain AUROC on TriviaQA.

Q1 – Existence

Do H-Neurons exist in smaller models?



Confirmed

H-Neuron classifiers outperform random baselines in all seven evaluated Pythia models.

This suggests that hallucination-associated FFN neurons are not exclusive to very large LLMs.

Q2 – Scale

Does performance improve with model size?



Mostly supported

Excluding caveat models, AUROC increases with scale:

$0.71 \rightarrow 0.80 \rightarrow 0.90 \rightarrow 0.92$

The 6.9B model slightly drops to 0.85; Pythia-1B is an outlier with only 3 H-Neurons.

Q3 – Generalization

Do H-Neurons transfer to NQ-Open?



Confirmed

Reliable models reach statistically significant OOD AUROC:

0.80 to 0.90

The 70M model could not produce a reliable OOD contrastive dataset.

- ▶ **Dataset construction is a bottleneck:** small models often fail to produce enough correct answers, requiring careful balancing.
- ▶ **Evaluation depends on answer normalization:** permissive aliases can create false positives if question text is repeated.
- ▶ **CETT is approximate here:** because quantized/offloaded models complicate access to all FFN weights, the project uses activation-normalized CETT.
- ▶ **Compute constraints shape methodology:** GPU memory, Colab availability, and checkpointing strongly influenced experimental design.

- ▶ **Causal perturbation experiments:** scale H-Neuron activations in Pythia models to test whether they reproduce the over-compliance pattern reported by Gao et al.
- ▶ **Cross-domain generalization:** evaluate transfer beyond NQ-Open, for example on biomedical QA datasets such as BioASQ.
- ▶ **Larger and more robust datasets:** expand the contrastive dataset to 500–1,000 examples per model and use stratified k -fold cross-validation.

Papers and background

- ▶ Gao et al. (2025), H-Neurons for hallucination detection.
- ▶ Transformer Circuits / mechanistic interpretability tutorials.
- ▶ Work on sparse probing and causal tracing in transformers.
- ▶ Background on hallucination in open-domain QA.

Practical starting path

- ▶ Start with a small open model such as Pythia-70M or 160M.
- ▶ Use TriviaQA or NQ-Open with short-answer prompts.
- ▶ Learn PyTorch forward hooks and Hugging Face model internals.
- ▶ Train a simple sparse logistic probe on internal activations.
- ▶ Always compare against random-neuron baselines.

Recommended tools: PyTorch, Hugging Face Transformers/Datasets, scikit-learn, TransformerLens-style interpretability notebooks, and careful checkpointing.

Thanks!